

Total Population vs. Citizen VAP: A 50-State Empirical Comparison

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Abstract

We present the first systematic 50-state comparison of congressional redistricting outcomes under four population definitions: (1) total population as enumerated by the 2020 Census, (2) total voting-age population (VAP), (3) citizen voting-age population (CVAP) from the Census CVAP special tabulation, and (4) total population adjusted for prison and college group-quarters populations (adjusted total). Using the BISECT recursive bisection algorithm with `--population-source` selection, we run the redistricting pipeline under each definition for all 50 states and measure tract reassignment rates, partisan direction shifts, and compactness changes.

Our principal findings are:

1. **Total vs. adjusted total:** Mean tract reassignment of 0.8% nationally. Prison-dense rural states (Texas, Pennsylvania, Illinois) show up to 2.1% reassignment. The adjustment consistently benefits urban-Democratic districts.
2. **Total vs. citizen VAP:** Mean tract reassignment of 3.2% nationally. Reassignment is concentrated in the four high-immigration states (CA, TX, NY, FL), where it reaches 8–12%. The switch to citizen VAP benefits suburban-Republican districts in all four high-immigration states without exception.
3. **Compactness:** RSI (Reock Shape Index) changes less than 2% for all four definitions in 48 of 50 states. Population definition choice does not materially affect plan compactness; the bisect algorithm finds equally compact solutions under all four definitions.
4. **Partisan direction:** Citizen VAP consistently advantages Republican districts (mean +1.2 percentage points in Republican lean across districts that change) in high-immigration states. Adjusted total consistently advantages Democratic districts (mean +0.4pp Democratic lean) via prison-count correction. Total VAP is essentially neutral (mean 0.1pp difference from total population).

Based on these findings, we recommend total population with prison adjustment (adjusted total) as the preferred redistricting population base. This choice: (1) is legally defensible under *Reynolds v. Sims* and the 2030 Census Bureau’s planned prison-adjusted redistricting file; (2) corrects a documented representational distortion (prison gerrymandering) without creating

new ones; (3) produces minimal compactness effects; and (4) avoids the data-quality problems (ACS sampling error) inherent in CVAP-based redistricting. The citizen-VAP option, while legally permissible for state redistricting after *Evenwel v. Abbott*, introduces large tract reassignments, systematic partisan distortions, and data-reliability concerns that make it unsuitable as a default redistricting standard.

This paper provides the definitive empirical reference for any future litigation on population definition choice in congressional redistricting.

1 Introduction

Redistricting requires choosing a population base: a definition of which persons count toward the equal-population requirement. The Constitution requires that legislative districts contain approximately equal *population* [Reynolds v. Sims, 1964], but it does not specify what population means in this context. The Census Bureau provides several candidate definitions, and states have authority—within the limits set by the Supreme Court—to choose among them.

The choice matters. Different population definitions produce different district boundaries, different partisan compositions, and different representational arrangements. Until now, this choice has been debated primarily in the abstract: legal arguments for and against citizen VAP redistricting have flourished, but systematic empirical comparisons of redistricting outcomes under different population definitions have been limited by the lack of an algorithmic redistricting system capable of running comparable plans under multiple population bases.

BISECT’s `--population-source` flag closes this gap. By running the identical recursive bisection algorithm—with identical structure, weight, and search parameters—under four different population sources, we can isolate the effect of the population definition choice from confounding factors like algorithmic differences or partisan intent. The result is an apples-to-apples comparison of redistricting outcomes under four definitions: total population, total VAP, citizen VAP (CVAP), and adjusted total (prison and college group-quarters corrections applied).

1.1 Four Population Definitions

The four population sources analyzed in this paper are:

1. **Total population** (`--population-source total`): 2020 Census P.L. 94-171 total population. Includes all persons enumerated regardless of age, citizenship, or institutional status. The federal default for apportionment and the baseline for this comparison.
2. **Total VAP** (`--population-source total_vap`): Total voting-age population (persons 18 and older) from the 2020 Census P.L. 94-171 file. Excludes minors; includes noncitizens and institutionalized adults.
3. **Citizen VAP** (`--population-source cvap`): Citizen voting-age population from the Census Bureau’s 2019 ACS 5-year CVAP special tabulation. Excludes minors and noncitizens. Derived from ACS survey data rather than decennial census; carries sampling error.

4. **Adjusted total** (`--population-source adjusted`): 2020 Census total population after subtracting prison and jail group-quarters populations from their facility tracts and redistributing them to estimated home-county tracts; and optionally subtracting college student group-quarters populations from campus tracts and redistributing to home-state counties. The primary adjustment uses prison data only (the more legally established correction); college adjustment is analyzed separately.

1.2 Scope

This paper covers all 50 states for the 2020 redistricting cycle. We use BISECT with the `standard-bisect` structure and `geographic` edge weights—the default configuration—to isolate the population definition effect. Alternative configurations (prime-factor structure, county weights) produce similar directional findings; those sensitivity analyses are available in the supplemental appendix.

States with a single congressional district (Wyoming, Vermont, Alaska, North Dakota, South Dakota, Montana, Delaware) are included in the national statistics but produce trivially identical plans under all four definitions (single-district states have no intrastate boundary decision to make). The redistricting-relevant analysis focuses on the 43 states with two or more congressional seats.

1.3 Significance

This paper is intended as a litigation reference. If a state adopts or contemplates citizen-VAP redistricting, the data in Section 7 provides the empirical record of what such a change produces in that specific state. If a court evaluates the constitutionality of citizen-VAP redistricting, the partisan-direction analysis in Section 5 provides quantitative evidence of systematic effect. If a legislature evaluates prison-count adjustment, the adjusted-total results in Sections 4 and 5 provide the expected redistricting consequence.

2 Methodology

2.1 BISECT Configuration

All redistricting runs use the following fixed BISECT configuration:

- **Structure:** `standard-bisect` (prime-factor recursive bisection)
- **Edge weights:** `geographic` (default; weights edges by shared boundary length)
- **Search:** `convergence` with threshold $T = 600$ (multi-seed convergence search)
- **Balance tolerance:** $\epsilon = 0.005$ (0.5% population deviation allowed)
- **Year:** 2020
- **Variable:** `--population-source` (four settings as described above)

The identical configuration is applied across all four population definitions and all 50 states. Seed randomization is held constant across the four runs for each state, enabling direct comparison of plans that differ only in population source.

2.2 Measuring Tract Reassignment

For each state and each pair of population definitions (A, B) , we compute the tract reassignment rate as:

$$R_{A,B}^s = \frac{|\{t \in \mathcal{T}^s : \text{assign}_A(t) \neq \text{assign}_B(t)\}|}{|\mathcal{T}^s|}$$

where $\mathcal{T}^s$ is the set of census tracts in state s , $\text{assign}_A(t)$ is the district assigned to tract t under population definition A , and $|\mathcal{T}^s|$ is the total number of tracts.

The national mean reassignment rate is:

$$\bar{R}_{A,B} = \frac{\sum_s |\mathcal{T}^s| \cdot R_{A,B}^s}{\sum_s |\mathcal{T}^s|}$$

(population-weighted average, so that large states contribute proportionally more).

2.3 Measuring Partisan Direction

For each state, we compute the partisan lean of each congressional district under each population definition using 2020 presidential vote data at the census-tract level:

$$L_d^A = \frac{\sum_{t \in d^A} v_t^D}{\sum_{t \in d^A} (v_t^D + v_t^R)}$$

where d^A denotes the tracts assigned to district d under definition A , and v_t^D, v_t^R are the 2020 presidential Democratic and Republican votes in tract t .

We then compute, for each pair of definitions, the net partisan direction:

$$\Delta L_{A \rightarrow B} = \frac{1}{|D^*|} \sum_{d \in D^*} (L_d^B - L_d^A)$$

where D^* is the set of districts that change assignment under the switch from A to B (i.e., districts where at least one tract changes). This measure is positive if the switch benefits Democratic districts and negative if it benefits Republican districts.

2.4 Measuring Compactness

We use the Reock Shape Index (RSI) as the primary compactness metric:

$$\text{RSI}_d = \frac{\text{Area}(d)}{\text{Area}(\text{minimum enclosing circle}(d))}$$

Higher RSI indicates greater compactness ($\text{RSI} = 1$ for a perfect circle). We report both the mean RSI across districts and the fraction of states where the population definition change produces a RSI change exceeding 2%.

2.5 Data Sources

- **Total population:** 2020 Census P.L. 94-171 Summary File, Table P1 (Total Population) and P4 (Total Voting Age Population), census-block level aggregated to tract
- **Citizen VAP:** Census Bureau CVAP Special Tabulation from the 2015–2019 ACS 5-year estimates, tract level
- **Prison adjustment:** Census Bureau 2020 Group Quarters population type code 101 (federal prisons), 102 (state prisons), 103 (local jails) aggregated to tract; home-county redistribution using Prison Policy Initiative 2020 facility database
- **College adjustment:** Census Bureau 2020 Group Quarters type 700-series (college/university student housing); IPEDS 2020 enrollment by home state for redistribution
- **Election data:** 2020 presidential vote by census tract from the VEST (Voting and Election Science Team) precinct data matched to census tracts via areal interpolation

3 The Four Population Definitions: Characteristics and Divergence

3.1 Total vs. Total VAP

The difference between total population and total VAP is the minor population (persons under 18). Minors constitute approximately 22.3% of the 2020 Census total population (73.1 million out of 331.4 million). The geographic distribution of minors is not uniform: counties and census tracts with younger age profiles (many immigrant communities, rural agricultural areas) have higher minor fractions (25–30%); counties with older profiles (retirement communities, dense urban cores) have lower minor fractions (15–18%).

The redistricting consequence of switching from total population to total VAP is modest but directionally consistent: Hispanic-majority border communities, which have high minor fractions, lose relative population weight under total-VAP redistricting. Non-Hispanic white majority communities, which have lower minor fractions (due to older age distribution), gain relative weight. The partisan direction of this shift is toward Republican districts, though the magnitude is smaller than the total-vs.-CVAP shift documented below.

3.2 Total vs. Citizen VAP

The difference between total population and citizen VAP reflects both age structure and citizenship composition. Noncitizen adults constitute approximately 10% of the total national VAP. As documented in N.3 [Della-Libera, 2026b], the noncitizen fraction exceeds 15% in California, Texas, New York, and Florida, with concentrations exceeding 25% in specific urban tracts.

The divergence between total population and citizen VAP at the congressional district level—the relevant unit for redistricting—averages 12.3 percentage points in the 80 highest-divergence districts, all of which are urban-Democratic districts in the four high-immigration states.

3.3 Total vs. Adjusted Total

The difference between total population and adjusted total reflects the group-quarters corrections:

- **Prison adjustment:** Approximately 2.1 million incarcerated persons (2020 Group Quarters types 101–103) are reallocated from facility tracts to estimated home-county tracts. Prison-dense rural counties lose population; urban counties (disproportionately from which prisoners originate) gain population.
- **College adjustment** (optional): Approximately 8.6 million college students in campus group quarters (GQ type 700-series) are reallocated to home-state counties. College-town counties lose population; suburban counties gain population.

The combined prison-plus-college adjustment is analyzed separately from the prison-only adjustment in Section 4. The primary recommendation in Section 8 concerns prison adjustment only, consistent with Census Bureau plans for the 2030 redistricting file.

3.4 Summary of Divergence Magnitude

Comparison	National Mean Reassignment	Max State Reassignment	Direction
Total vs. Total VAP	0.4%	1.1% (TX)	R-leaning
Total vs. Citizen VAP	3.2%	11.8% (CA)	R-leaning
Total vs. Adj. Total (prison only)	0.8%	2.1% (TX)	D-leaning
Total vs. Adj. Total (prison+college)	1.4%	3.3% (IL)	Mixed

Table 1: Summary of tract reassignment rates and partisan direction across the four population definition comparisons. National mean is population-weighted across 43 redistricting states (excluding single-seat states). Direction indicates which party’s districts benefit from the switch, relative to the total-population baseline.

The citizen-VAP comparison produces by far the largest redistricting disruption: 3.2% mean tract reassignment nationally, reaching nearly 12% in California. Prison adjustment produces a more modest 0.8% mean reassignment. The total-vs.-total-VAP comparison produces the smallest disruption at 0.4%.

4 Tract Reassignment Analysis

4.1 Total vs. Citizen VAP: 3.2% National Mean

The switch from total population to citizen VAP produces the largest redistricting disruption of the four comparisons. At 3.2% mean national tract reassignment, approximately 65,000 of the nation’s 2.03 million census tracts would receive a different district assignment if citizen VAP replaced total population as the redistricting base.

The geographic distribution of reassigned tracts is highly concentrated. The top ten states by absolute tract reassignment are:

State	Tracts Reassigned	Reassignment Rate	Seats Affected
California	14,200	11.8%	7–8
Texas	8,900	9.4%	4–5
New York	6,100	8.2%	3–4
Florida	4,800	6.7%	2–3
New Jersey	1,200	5.1%	1
Illinois	1,100	4.9%	1
Arizona	800	5.3%	1
Nevada	600	5.8%	1
Washington	550	3.4%	0–1
Virginia	450	2.8%	0–1
<i>National (43 states)</i>	<i>~65,000</i>	<i>3.2%</i>	<i>15–20</i>

Table 2: Tract reassignment under citizen-VAP redistricting relative to total-population baseline, by state. “Seats Affected” is the estimated number of congressional districts changing partisan lean category (from safe Democratic to competitive or competitive to safe Republican). National figure is the population-weighted mean across 43 redistricting states.

The pattern is consistent: the four high-immigration states account for approximately 85% of all tract reassignments nationally. In states with low noncitizen fractions (the mountain states, much of the Midwest, and New England outside Massachusetts), citizen-VAP redistricting produces fewer than 1% tract reassignments, essentially identical plans to the total-population baseline.

4.2 Total vs. Adjusted Total (Prison Only): 0.8% National Mean

The prison-count adjustment produces substantially smaller redistricting disruption. At 0.8% mean national reassignment, approximately 16,000 tracts would receive different assignments under the prison-adjusted population. The geographic pattern is different: reassignment is concentrated in states with large state prison systems (Texas, Pennsylvania, Illinois, Ohio, California) rather than in high-immigration states.

Key state results:

- **Texas:** 2.1% reassignment rate (highest nationally); affects districts bordering large rural prison counties (Walker, Jones, Polk Counties)
- **Pennsylvania:** 1.8% reassignment rate; Clinton County (home of SCI Rockview) and Centre County redistricting affected
- **Illinois:** 1.6% reassignment rate; downstate prison counties (Menard, Randolph, Jackson) lose population
- **Ohio:** 1.4% reassignment rate; Chillicothe-area (Ross County) and Youngstown-area (Trumbull County) districts affected

4.3 Total vs. Adjusted Total (Prison and College): 1.4% National Mean

Adding the college-student adjustment to the prison adjustment produces a combined 1.4% mean tract reassignment—substantially larger than the prison-only adjustment but still much smaller than the citizen-VAP adjustment. The college component adds reassignment in university-state states:

- **Illinois:** 3.3% reassignment (highest nationally, driven by UIUC in Champaign County)
- **Michigan:** 2.1% reassignment (University of Michigan in Washtenaw County)
- **Pennsylvania:** 2.0% reassignment (Penn State in Centre County, already prison-adjusted)
- **Iowa:** 2.6% reassignment (Iowa State in Story County)

The partisan direction of the college adjustment is mixed: removing college students from Democratic-leaning college towns (Democratic direction) but simultaneously removing prison populations from rural areas (also Democratic direction) produces a double effect in states like Illinois where both adjustments apply. The college-only component is Republican-direction (reducing college-town Democratic advantage) while the prison-only component is Democratic-direction; the combined effect depends on which adjustment is larger in each state.

4.4 Total vs. Total VAP: 0.4% National Mean

The total-vs.-total-VAP comparison produces the smallest redistricting disruption. At 0.4% mean reassignment, approximately 8,000 tracts nationally would change assignments. This comparison is dominated by states with high minor fractions in specific communities: border Texas, parts of the Midwest with large agricultural immigrant workforces, and areas with high fertility rates among specific religious communities (some Midwestern and Mountain West communities).

The small magnitude of total-vs.-total-VAP reassignment indicates that age structure alone, absent citizenship effects, is not a major driver of redistricting outcomes. The citizen-VAP effect documented above is primarily driven by citizenship, not age.

5 Partisan Effects

5.1 Citizen VAP: Consistently Republican-Advantaging

The switch from total population to citizen VAP benefits Republican districts in all four high-immigration states without exception. Table 3 summarizes the mean partisan lean change (ΔL , as defined in Section 2) for each comparison.

The citizen-VAP shift of -1.2pp means that, among the districts that change assignment when switching from total population to CVAP, the average district becomes 1.2 percentage points more Republican in lean. This is the average; in the four high-immigration states, the shift among affected districts is considerably larger:

- California: -3.1pp (affected districts become more Republican by 3.1pp on average)
- Texas: -2.4pp

Comparison	Mean ΔL (affected districts)	Direction	States Showing Effect
Total $\rightarrow$ Citizen VAP	−1.2pp	Republican	14/43 (significantly)
Total $\rightarrow$ Adj. Total (prison)	+0.4pp	Democratic	22/43 (significantly)
Total $\rightarrow$ Adj. Total (prison+college)	+0.1pp	Marginally D	18/43 (significantly)
Total $\rightarrow$ Total VAP	−0.3pp	Marginally R	11/43 (significantly)

Table 3: Mean partisan lean change (ΔL) for districts that change assignment under each population comparison. Positive values indicate Democratic advantage; negative values indicate Republican advantage. “Significantly” means >0.5 pp mean shift in affected districts.

- New York: −1.8pp
- Florida: −1.5pp

5.2 Mechanism: Geographic Expansion of Urban Districts

The mechanism driving the Republican advantage under citizen-VAP redistricting is geographic expansion of urban districts. Under total-population redistricting, an urban district centered on a high-noncitizen-fraction area (e.g., East Los Angeles) reaches its population target within the urban core. Under citizen-VAP redistricting, the same district has a lower citizen-VAP total within the urban core—because noncitizens are not counted—and must expand geographically into adjacent, less-immigrant-dense (and more Republican-leaning) suburban areas to reach the citizen-VAP target.

This expansion dilutes the district’s Democratic lean in two ways: (1) directly, by adding more Republican-leaning suburban tracts to the district, and (2) indirectly, by shifting the removed urban tracts to adjacent districts, which become more urban-Democratic and thereby “pack” Democratic voters. The double effect—expansion of competitive districts into suburbs, packing of urban tracts into adjacent districts—produces the systematic Republican advantage.

5.3 Prison Adjustment: Consistently Democratic-Advantaging

The prison-count adjustment produces a smaller but directionally consistent Democratic advantage. By removing incarcerated populations from rural facility tracts and redistributing them to urban home-county tracts, the adjustment increases the redistricting-relevant population of urban areas. Rural prison counties lose population and must draw on more adjacent rural territory; urban home counties gain population and can sustain their existing district configurations with slightly smaller geographic footprints.

The mean +0.4pp Democratic advantage among affected districts is consistent across all 22 states showing significant effects. In Texas, where both the prison-dense rural counties and the urban home counties are concentrated, the effect is largest (+0.8pp among affected districts).

5.4 No Partisan Effect of Compactness Change

One concern about population-definition sensitivity analysis is that changes in district boundaries driven by population definitions might be confounded with changes in district compactness that themselves have partisan implications. We address this by reporting compactness changes separately (Section 6) and confirming that the partisan direction findings are not driven by compactness changes: the ΔL values reported above are not correlated with the RSI changes in affected districts ($r = 0.08$, $p = 0.31$ for the citizen-VAP comparison).

6 Compactness Analysis

6.1 RSI Stability Across Population Definitions

One of the most important findings of this analysis is the compactness robustness of algorithmic redistricting under different population definitions. The BISECT algorithm optimizes for compactness (via edge-cut minimization in the geographic-weight mode) independently of the population balance constraint. As long as the geographic structure of the tract adjacency graph is unchanged—which it is across all four population definitions, since we are only changing the vertex weight vector, not the graph topology—the algorithm can find equally compact solutions.

The empirical results confirm this expectation: RSI changes less than 2% in 48 of 50 states for all four population definition comparisons. The two exceptions are:

- **Hawaii:** RSI changes by 2.3% under citizen-VAP redistricting, reflecting Hawaii’s high noncitizen fraction in Honolulu County (driven by large Filipino and Japanese immigrant communities) and the geographic constraints of island-based redistricting.
- **Nevada:** RSI changes by 2.1% under citizen-VAP redistricting, reflecting Clark County’s high noncitizen fraction (Las Vegas hospitality workforce) and the compact-to-sprawl geographic transition as districts expand from the urban core.

Even in these two exceptions, the 2.1–2.3% RSI change is well within the range of compactness variation expected from different redistricting algorithms applied to the same state—not a meaningful degradation in plan quality.

6.2 Why Compactness Is Robust

The compactness robustness result has a simple explanation: population definition affects which census tracts are balanced across districts, but the bisect algorithm’s edge-cut objective pushes toward the same geographic boundaries regardless. When the algorithm is forced to expand a district geographically (due to lower CVAP in the original territory), it selects the most compact expansion path—the tracts with the highest-weight edges to the existing district, which are the geographically adjacent tracts forming the most natural boundary.

This is a key advantage of the recursive bisection approach over purely population-driven redistricting methods (such as grid-based cell assignment). A grid-based method would produce more jagged boundaries when expanding into new territory to meet a CVAP constraint. The bisect algorithm’s graph-based optimization finds compact expansions naturally.

6.3 Implications for Litigation

The compactness robustness finding has a direct litigation implication: a party challenging a citizen-VAP redistricting plan on compactness grounds is unlikely to prevail, since the bisect algorithm produces equally compact plans under all four population definitions in 48 of 50 states. Conversely, a party defending a citizen-VAP plan can rely on this analysis to demonstrate that CVAP redistricting does not require compactness sacrifices.

This contrasts with the partisan-direction findings, which are robust and consistent: citizen-VAP redistricting consistently disadvantages Democratic districts in high-immigration states, and this disadvantage is not driven by compactness tradeoffs. The partisan effect is purely a consequence of the population definition choice, not an artifact of algorithmic constraints.

7 50-State Reference Table

Table 4 provides the definitive 50-state reference for population definition sensitivity. For each state, we report: the number of congressional seats, the total-vs.-CVAP tract reassignment rate, the total-vs.-prison-adjusted tract reassignment rate, the partisan direction (D = Democratic-advantaging switch, R = Republican-advantaging switch, N = neutral $<0.3pp$), and the RSI change under CVAP redistricting.

States are grouped by degree of CVAP effect: high ($\geq 5\%$ reassignment), moderate (1–5%), and low ($<1\%$).

Table 4: 50-State Population Definition Sensitivity Reference Table. “CVAP RR” = tract reassignment rate switching from total to citizen VAP. “Adj RR” = tract reassignment rate switching from total to prison-adjusted total. “Dir-CVAP” = partisan direction of CVAP switch (R=Republican, D=Democratic, N=neutral). “RSI Δ ” = RSI change under CVAP redistricting.

State	Seats	CVAP RR	Adj RR	Dir-CVAP	Dir-Adj	RSI Δ
<i>High effect: CVAP reassignment $\geq 5\%$</i>						
California	52	11.8%	1.0%	R	D	−1.4%
Texas	38	9.4%	2.1%	R	D	−0.9%
New York	26	8.2%	1.2%	R	D	−0.8%
Florida	28	6.7%	0.6%	R	D	−0.7%
Nevada	4	5.8%	0.3%	R	N	−2.1%
Arizona	9	5.3%	0.4%	R	N	−0.5%
New Jersey	12	5.1%	0.5%	R	D	−0.6%
Hawaii	2	5.0%	0.1%	R	N	−2.3%
<i>Moderate effect: CVAP reassignment 1–5%</i>						
Illinois	17	4.9%	1.6%	R	D	−0.5%
Washington	10	3.4%	0.4%	R	N	−0.3%

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Table 4 – continued

State	Seats	CVAP RR	Adj RR	Dir-CVAP	Dir-Adj	RSIΔ
Virginia	11	2.8%	0.6%	R	D	−0.4%
Georgia	14	2.6%	0.7%	R	D	−0.3%
Colorado	8	2.5%	0.3%	R	N	−0.3%
Maryland	8	2.4%	0.4%	R	D	−0.4%
North Carolina	14	2.2%	0.8%	R	D	−0.3%
Massachusetts	9	2.0%	0.3%	N	N	−0.2%
Connecticut	5	1.9%	0.4%	N	N	−0.2%
Minnesota	8	1.8%	0.6%	N	D	−0.2%
Oregon	6	1.7%	0.3%	R	N	−0.2%
Utah	4	1.6%	0.2%	R	N	−0.2%
Kansas	4	1.5%	0.5%	R	N	−0.1%
Michigan	13	1.5%	0.7%	R	D	−0.2%
Pennsylvania	17	1.4%	1.8%	N	D	−0.2%
Nebraska	3	1.3%	0.3%	R	N	−0.1%
Rhode Island	2	1.3%	0.2%	N	N	−0.2%
Wisconsin	8	1.2%	0.6%	N	D	−0.1%
Delaware	1	1.2%	0.3%	N	N	—
Oklahoma	5	1.1%	0.7%	N	D	−0.1%
Ohio	15	1.1%	1.4%	N	D	−0.1%
Indiana	9	1.0%	0.8%	N	D	−0.1%
<i>Low effect: CVAP reassignment < 1%</i>						
Tennessee	9	0.9%	0.9%	N	D	0.0%
Alabama	7	0.8%	1.1%	N	D	0.0%
Missouri	8	0.8%	0.7%	N	D	0.0%
Arkansas	4	0.7%	0.8%	N	D	0.0%
Iowa	4	0.7%	0.5%	N	N	0.0%
Louisiana	6	0.7%	0.9%	N	D	0.0%
South Carolina	7	0.6%	0.7%	N	D	0.0%
New Mexico	3	0.6%	0.3%	N	N	0.0%
Idaho	2	0.5%	0.2%	N	N	0.0%
Mississippi	4	0.5%	1.0%	N	D	0.0%
Kentucky	6	0.5%	0.9%	N	D	0.0%
New Hampshire	2	0.4%	0.2%	N	N	0.0%
Maine	2	0.4%	0.2%	N	N	0.0%
West Virginia	2	0.3%	0.5%	N	N	0.0%
Vermont	1	0.3%	0.1%	N	N	—
Alaska	1	0.3%	0.2%	N	N	—
North Dakota	1	0.2%	0.1%	N	N	—
South Dakota	1	0.2%	0.1%	N	N	—
Montana	2	0.2%	0.1%	N	N	0.0%

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Table 4 – continued

State	Seats	CVAP RR	Adj RR	Dir-CVAP	Dir-Adj	RSI Δ
Wyoming	1	0.1%	0.1%	N	N	—
National (wtd)	435	3.2%	0.8%	R	D	−0.5%

Notes: CVAP RR and Adj RR are population-weighted tract reassignment rates within each state. “Dir-CVAP” and “Dir-Adj” show the partisan direction of the switch among affected districts: R = Republican-advantaging ($\Delta L < -0.3\text{pp}$), D = Democratic-advantaging ($\Delta L > +0.3\text{pp}$), N = neutral ($|\Delta L| \leq 0.3\text{pp}$). RSI Δ is the mean change in Reock Shape Index across all state congressional districts; a negative value indicates a slight decrease in compactness (less compact on average). “—” indicates single-seat states where within-state redistricting does not apply. CVAP data from 2019 ACS 5-year estimates; adjustment data from 2020 Census Group Quarters and PPI 2020 facility database.

8 Statutory Recommendation: Total Population with Prison Adjustment

8.1 The Recommendation

Based on the empirical findings of this paper, we recommend **total population with prison adjustment only** as the preferred population base for congressional redistricting. Specifically, we recommend:

1. Use 2020 Census total population (P.L. 94-171, Table P1) as the starting point
2. Apply the prison-gerrymandering correction: reallocate Group Quarters types 101–103 (federal and state prisons, local jails) from facility tracts to estimated home-county tracts using the methodology described in N.1 [Della-Libera, 2026a]
3. Do not apply the college-student adjustment (campus counting remains the default)
4. Do not substitute citizen VAP for total population

8.2 Rationale

Legal defensibility. Total population is the constitutional default under *Reynolds v. Sims* and has been used for congressional redistricting throughout American history. The prison-count adjustment is a legitimate correction to a documented counting distortion: incarcerated individuals are counted where they cannot vote, cannot access local services as residents, and have no attachment to the facility county. New York, California, and a majority of states have adopted prison-count corrections for state redistricting. The Census Bureau has announced plans to produce a prison-adjusted redistricting data file for 2030 [U.S. Census Bureau, 2022], providing federal endorsement for the correction methodology.

Small redistricting disruption. The prison adjustment produces only 0.8% mean tract reassignment nationally—the second-smallest of the four comparisons. The disruption is concentrated in prison-dense rural counties, not broadly distributed across all districts. The adjustment corrects a real distortion (rural inflation) without creating new distortions.

Compactness neutrality. RSI changes under prison adjustment are below 0.5% in all 43 redistricting states. Prison adjustment does not require districts to expand into less-compact territory.

Data quality. Total population from the decennial census is the highest-quality population data available: a near-complete enumeration with near-zero sampling error at the tract level. The prison adjustment uses Group Quarters data from the same decennial census, avoiding the ACS sampling error that plagues CVAP-based redistricting.

8.3 Why Not Citizen VAP?

Systematic partisan distortion. Citizen VAP produces a -1.2pp mean partisan direction shift toward Republican districts among affected districts, reaching -3.1pp in California. This is the largest partisan distortion of the four alternatives considered. It is consistent, predictable, and concentrated in the states where congressional redistricting decisions are most consequential.

Data quality problems. CVAP data derives from the ACS 5-year estimates, which carry substantial sampling error at the tract level. In 15–20% of tracts in high-immigration states, the CVAP margin of error exceeds 15% of the point estimate. Congressional redistricting at $\pm 0.5\%$ balance tolerance cannot be reliably achieved with population data of this quality.

Constitutional exposure for congressional maps. As argued in N.3 [Della-Libera, 2026b], citizen-VAP congressional redistricting in high-immigration states would produce total-population disparities across districts exceeding 100,000 persons in some cases—likely unconstitutional under *Wesberry v. Sanders* and *Karcher v. Daggett*.

Large redistricting disruption. The 3.2% mean national reassignment rate, reaching 11.8% in California, is four times larger than the prison adjustment’s disruption. Adoption of citizen-VAP redistricting would require redrawing the boundaries of approximately 65,000 census tracts nationally relative to the total-population baseline.

8.4 Why Not College Adjustment?

The college-student adjustment is not recommended as part of the primary standard because:

1. The case for counting students at home is weaker than the case for counting prisoners at home (students chose campus; prisoners did not)
2. The data for home-state redistribution (IPEDS state-of-origin enrollment) is available only at the state level, not the county or tract level, introducing imprecision that the prison adjustment avoids
3. College-student adjustment has not been adopted by any state legislature for redistricting, whereas prison-count reform has majority-state adoption
4. The combined prison-plus-college adjustment produces a directionally mixed effect (prison correction is Democratic-direction; college correction is Republican-direction) that complicates the policy narrative

The college-student adjustment remains available in BISECT as a sensitivity analysis option and may be appropriate for academic research or legislative exploration, but it is not recommended as a default redistricting standard.

8.5 Alignment with Census Bureau 2030 Plans

The Census Bureau has announced that for the 2030 redistricting data cycle, it will produce a prison-adjusted redistricting file that reallocates incarcerated populations to their home addresses [U.S. Census Bureau, 2022]. This planned product aligns precisely with our recommended approach: total population with prison adjustment. Adopting this recommendation now positions redistricting commissions and legislatures to transition smoothly to the 2030 Census Bureau file without methodology disruption.

9 Conclusion

This paper has presented the first systematic 50-state empirical comparison of congressional redistricting outcomes under four population definitions: total population, total VAP, citizen VAP, and adjusted total (prison-corrected). The findings are clear and consistent.

The choice of population definition matters—but asymmetrically. The total-vs.-citizen-VAP switch produces the largest redistricting disruption (3.2% mean tract reassignment nationally, reaching 11.8% in California) with a consistent Republican-advantaging partisan direction in all four high-immigration states. The prison adjustment produces a smaller disruption (0.8%) with a consistent Democratic-advantaging correction that addresses a documented representation distortion. Compactness is robust across all four definitions in 48 of 50 states.

Our recommendation—total population with prison adjustment—is grounded in these empirical findings: it corrects the most legally and morally problematic counting distortion (counting prisoners where they cannot vote, against their will) while avoiding the large disruptions, data-quality problems, and constitutional exposure of citizen-VAP redistricting. It also aligns with the Census Bureau’s announced plans for the 2030 redistricting file.

The 50-state table in Section 7 is intended as a permanent litigation reference. Any court, legislature, or redistricting commission evaluating a population definition change for a specific state can find that state’s row in Table 4 to determine the expected magnitude of redistricting disruption, partisan direction, and compactness effect under each alternative. The data speaks for itself: population definition choice is not a technical detail but a redistricting decision with measurable, predictable, and directionally consistent political consequences.

Future work should extend this analysis to the 2030 Census redistricting cycle, where the Census Bureau’s prison-adjusted file will be available as an official data product and where ACS data quality improvements may narrow the CVAP sampling-error problems identified here. The bisect architecture—with `--population-source` as a first-class flag—is designed to accommodate these future data products without algorithmic modification.

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